MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 15: Tuesday 06 October 2026

(Week 8)

Remark. Q15.1 to Q15.8 are on Sections 14.1, 14.2, 14.3 of Thomas' Calculus.

Q15.1. Evaluate the double integral $\iint_R (x^4 + y^4)\, dA$, where R is the region in the xy-plane bounded by the lines $x = 0$, $x = 1$, $y = 0$ and $y = 2$.

Q15.2. Evaluate the double integral $\iint_R \frac{xy}{\sqrt{9 - x^2}}\, dA$, where R is the region in the xy-plane bounded by the lines $x = 0$, $x = \sqrt{5}$, $y = 1$ and $y = 5$.

Q15.3. Find the volume of the solid region under the graph of the function $f(x, y) = 3 + (x - 1)^2 + 4y^2$ and above the rectangular region $R : 0 \leq x \leq 3,\ \ 0 \leq y \leq 2$ in the xy-plane.

[Remark. Compute the volume as a double integral.]

Q15.4. Evaluate the iterated integral

$$\int_{1/4}^{1} \int_{y^4}^{y^2} \sqrt{\frac{y}{x}}\, dx\, dy.$$

Q15.5. Evaluate the double integral

$$\iint_D y^2 \cos x\, dA,$$

where D is the region bounded by the line $x = 0$, the curve $x = y^3$ and the line $y = 1$.

Q15.6. Evaluate the double integral $\iint_D (x + 4)\, dA$, where D is the region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$.

[Remark: You need to find the points of intersection of the parabola and the line.]

Q15.7. Evaluate the following integral by reversing the order of integration:

$$\int_0^{27} \int_{\sqrt[3]{y}}^{3} \left(\frac{1}{81} + e^{x^4}\right) dx\, dy.$$

Q15.8. Evaluate the following integral by reversing the order of integration:

$$\int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y}\, dy\, dx.$$